Mohamad **Dikshie Fauzie**

Principal Infrastructure Architect · Low-Latency & Distributed Systems

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Summary

Principal Infrastructure Architect and Systems Engineer with 10+ years of mission-critical production experience co-managing low-latency infrastructure at Japan's largest alternative trading venue (PTS), backed by a Ph.D. from Keio University (supervised by Internet Hall of Fame pioneer Prof. Jun Murai and quantum networking pioneer Prof. Rodney Van Meter) and an ITB Physics degree. Proven record delivering zero-downtime datacenter migrations, 50% rack power balancing optimizations, 50% storage throughput gains via Dell PowerScale (Isilon), and end-to-end telemetry automation. Combines deep physical-layer discipline (power/cooling capacity planning, optical metrology) with advanced systems engineering: sub-nanosecond precision timing (NPL Time & Frequency), low-latency TCP/UDP kernel tuning, NetFPGA, AI cluster infrastructure (NVIDIA certified), and cloud-native virtualization.

Skills

Systems & OS	Low-Latency Architectures, High-Performance Systems (HPC), AI/GPU Clusters, Linux (RHEL, Ubuntu), FreeBSD, Kernel Tuning
Networking	TCP/UDP, BGP, OSPF, IPv6 Core, IEEE 1588v2 PTP, Sub-Nanosecond Timing, SDN, P2P/Mesh Topologies, CDN, NetFPGA
Storage & Cloud	Dell PowerScale (Isilon) Storage, LTO Tape Media Backup, Kubernetes, Docker, OpenStack, ONAP, OPNFV, AWS Cloud
Facilities & DC	Rack PDU Power Telemetry & Balancing, Facility Thermal/Capacity Planning, Server Migration, Optical Metrology (SFPs/QSFPs)
Automation	Python, Bash/Shell, Git, NetBox (DCIM), SNMP Telemetry, Open-Source Monitoring & Inventory Systems, LaTeX
Languages	English (Fluent / Professional), Japanese (Working Proficiency), Indonesian (Native)

Work Experience

Japannext Co., Ltd. (Largest Proprietary Trading System / PTS in Japan)

Tokyo, Japan

Data Center Engineer (Co-Manager) / Infrastructure Lead — *Oct. 2014 - Present*

- Co-managed 24/7 mission-critical, ultra-low-latency datacenter operations for Japan's primary alternative equity trading venue, maintaining 99.999% infrastructure availability under FSA regulatory compliance.
- Led end-to-end datacenter server migration for power-constrained production systems; orchestrated risk mitigation protocols and executed with minimal downtime, delivering 2 weeks ahead of schedule and unlocking 50% expansion capacity.
- Spearheaded rack-level PDU power telemetry across high-density server racks, achieving a 50% improvement in workload power balancing and optimizing facility power and cooling capacity planning.
- Architected enterprise scale-out storage on Dell PowerScale (Isilon) and automated LTO tape media backup systems, achieving a 50% increase in server throughput and securing long-term compliance retention.
- Designed and deployed open-source DCIM inventory tracking (NetBox) for physical datacenter hardware, racks, and optical consumables (SFPs/QSFPs, fiber), drastically accelerating failure discovery and MTTR.
- Partnered with Unix infrastructure engineering to integrate an open-source hardware discovery platform with automated system monitoring, streamlining asset discovery, fault detection, and capacity planning.
- Automated executive monthly capacity, power, and SLA reporting using open-source telemetry pipelines, eliminating manual reporting overhead and boosting operational transparency.
- Directed datacenter room buildouts and server rack installations for critical internal trading systems and client co-location racks, achieving full operational readiness one week ahead of schedule.

Keio University - JICA JST Indo Project

Tokyo, Japan

Research Assistant (Distributed Systems & Networks) — *Apr. 2011 - Sep. 2014*

- Researched distributed network topologies, energy efficiency, and power consumption in large-scale peer-to-peer (P2P) networks and Content Delivery Networks (CDNs).
- Applied game theory models to information dissemination using probabilistic gossip protocols, analyzing graph dynamics in dynamic swarms.
- Published peer-reviewed research papers in international IEEE and IEICE journals and conference proceedings.

Keio University

Tokyo, Japan

System Administrator (Campus Core Infrastructure) — *Oct. 2007 - Sep. 2014*

- Administered campus server and core network infrastructure (DNS, email, web servers, BGP/OSPF routers, and switches) in coordination with lab managers, unifying administrative workflows for high operational reliability.

Education

Keio University — *Tokyo, Japan*

Ph.D. in Media and Governance (Internet Architecture & Distributed Systems) — *Graduated Oct. 2014*

- Dissertation: Dynamic P2P network topologies, swarm dynamics, and energy-efficient Content Delivery Networks (CDNs).
- Supervisors: Prof. Jun Murai (Internet Hall of Fame / WIDE Project), Prof. Rodney Van Meter (Quantum Networking Pioneer), Dr. Kenjiro Cho, Prof. Hiroshi Kusumoto.

Institut Teknologi Bandung (ITB) — *Bandung, Indonesia*

M.T. (Master of Technology in Electrical Engineering) — *Graduated Oct. 2004*

- Designed and benchmarked one of Indonesia's earliest IPv6 campus network backbones and core routing deployments.

Institut Teknologi Bandung (ITB) — *Bandung, Indonesia*

S.Si. (Bachelor of Science in Physics) — *Graduated Oct. 2000*

- Researched data assimilation methods in 1D hydrodynamic fluid systems (numerical modeling, wave equations, fluid mechanics).

Publications

Energy Consumption in Peer Assisted CDN

International Journal of Advanced Research in Computer Science (IJARCS) Vol.4 No.9

Lead Author — *Aug. 2013*

- Modeled and evaluated energy efficiency of peer-assisted edge CDN architectures versus traditional centralized distribution.

Assessing The Dynamics of BitTorrent Swarm Topologies Using The Peer Exchange Protocol

IEICE Transactions on Communications (Special Issue on Information Network Science, Vol.E95-B, No.05)

Lead Author — *2012*

- Empirically inferred dynamic overlay topologies and peer discovery mechanisms using the Peer Exchange (PEX) protocol.

A Temporal View of The Topology of Dynamic BitTorrent Swarms

IEEE INFOCOM Workshop on Network Science for Communication Networks (DOI: 10.1109/INFCOMW.2011.5928939)

Lead Author — *2011*

- Analyzed temporal graph evolution and topological robustness of large-scale distributed swarms under dynamic churn.

Certifications & Specialized Training

2024	**AI Infrastructure and Operations Fundamentals**, NVIDIA (Deep Learning Institute)	*Online*
2024	**Foundations of Project Management**, Google Career Certificates	*Online*
2022	**Introduction to Time and Frequency Measurement**, National Physical Laboratory (NPL), UK	*Teddington, UK*
2020	**Introduction to Metrology (Measurement Standards)**, National Physical Laboratory (NPL), UK	*Teddington, UK*
2010	**Introduction to NetFPGA (High-Speed Packet Processing)**, Stanford University	*Tokyo, Japan*
2023	**AWS Certified Cloud Practitioner**, Amazon Web Services (AWS)	*Online*
2022 - 2023	**Machine Learning & Mathematics for Machine Learning**, Stanford University / DeepLearning.AI	*Online*
2019	**Cloud Infrastructure, Kubernetes & OpenStack**, The Linux Foundation	*Online*
2019	**Network Transformation: 5G, Core Network, Cloud & NFV**, Intel	*Online*
2019	**Carrier Network Automation: ONAP, OPNFV & Open Source Networking**, The Linux Foundation	*Online*
2022	**Financial Risk Management & Capital Markets**, New York Institute of Finance / Interactive Brokers	*Online*